

David (wizzense)

Platform Engineer. Founder & Architect, Aitherium / AitherOS. USAF veteran.

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SUMMARY

I build paved roads. For 14 months I have run AitherOS - a 378-service platform where 33 agents plan, code, review, deploy and document their own work - and the only reason that is safe is the platform underneath it: 798 self-testing quality gates, ring-based progressive rollout with health-gated batches and one-command rollback, a 4-hourly in-fleet probe that pages on VIOLATION and on CANNOT-JUDGE separately, and a written doctrine that says a mechanically detectable defect must become a check, never a ticket. Before that, 12 years of enterprise infrastructure: the second-largest Active Directory on earth (850,000 users, 690 domain controllers) for the US Air Force, HPC tape and SAN at Boeing, and six years at Tanium as the escalation engineer customers got when the normal answers ran out. I have shipped 24,661 commits with no product manager, no delivery manager and nobody prodding me for status - and I can explain precisely how autonomous systems lie to you about their own health.

EXPERIENCE

Aitherium — Founder & Platform Architect (AitherOS, the aw* stack, awnix)

Jul 2025 - Present · Remote

- Own a 378-service platform end to end - podman quadlets on Rocky/Debian, Postgres with streaming replication, Redis Sentinel, Cloudflare Workers/Tunnels/Pages, k3s, self-hosted GitHub Actions runner fleets, and GPU scheduling across local, DGX and rented (AWS, Lambda, Vast, Hetzner) capacity.
- Built the guardrail layer: 798 self-testing checkers, each required to carry a --self-test proving it can still fail, each wired to a named unattended lane by a manifest gate that refuses a checker running nowhere. Exit codes are 0 clean / 1 violation / 2 could-not-judge, and could-not-judge never reads as a pass.
- Progressive delivery: ring-based deploy with health assertions, one-command promote and rollback, health-gated batch roll-recreate with the self-heal watchdog paused for the window, plus autoheal and a death watchdog for wedged services. A deploy-invariants checker measures which trees are baked into an image versus bind-mounted, and whether the process actually re-imported the change.
- Ran incident response on my own platform and converted each finding into a gate: a paging-path gate that round-trips the alert, added after detection worked and the page silently died; a client that retried on absence as if it were contention; an authz grant scope where read silently covered readwrite; a name-shaped role check that let a shop admin DELETE workspace nodes.
- Shipped 24,661 commits since June 2025 (16,393 in the last 90 days) across 34,206 tracked files, including 1.37M lines of TypeScript over 38 React and 16 Next.js surfaces, with no PM and no delivery manager.
- Wrote and enforced the written operating doctrine the team works from: rule files that load on demand, a 150-word fixed-order report format, and a ledger protocol where a mechanically detectable defect must become a check rather than a row - instituted after measuring the ledger at 79 rows/day in against 4.2/day out.
- Published four packages on PyPI (awdk, awgit, awgraph, awrelay) and 61 public repos; sold and operate three tenant appliances (repair shop, game studio, campaign tooling), each a generated repo, image, brain pack and portal the customer owns.

Tanium — Senior Enterprise Services Engineer (prev. Escalation Engineer, Senior TAM)

Jan 2022 - Present · Remote

- The engineer enterprise and federal customers reached when the normal answers ran out: platform SME on the hardest escalations, 195 cases in six months at 100% CSAT.
- Rewrote the Risk Assessment tooling in Python to move it off a 500-endpoint ceiling to unbounded; applied it to a 50,000-endpoint estate.
- Built Client Health dashboards and remediation automation that moved customers toward a self-healing endpoint fleet - the detect-to-action loop, not just the dashboard; wrote the guide and ran the live webinar to hundreds of attendees.
- 2x Star Performer (2024).

Boeing — HPC Linux System Engineer

Apr 2020 - Jan 2022 · Seal Beach, CA

- Owned the enterprise tape archive (Oracle HSM / SAM-QFS, then technical lead on the migration to Versity Storage Manager) and the fibre-channel SAN behind HPC workloads.
- Wrote the Python automation for recovery, monitoring and archival; led the LTO6 to LTO8 migration and the data-at-rest encryption rollout (OKM 3, IBM SKLM) without an archive outage.

United States Air Force - Cyber Systems Operations — Enterprise Active Directory Engineer; Functional Lead; Supervisor, Directory Services Operations

Jan 2014 - Jan 2020 · Honolulu HI; Lajes Field, Portugal; Keesler AFB

- Led Tier-2 operations for a \$10B, 850,000-user, 690-domain-controller, 230-site Active Directory - the second-largest AD network in the world - including classified and unclassified directory and Exchange services, 24/7, follow-the-sun.
- Performed an authoritative ADDS restore recovering 3,000 deleted accounts with zero collateral impact - the highest-stakes recovery I have run, on a system where a wrong replication decision propagates globally in minutes.
- Automated maintenance and health checks for 70+ domain controllers in PowerShell, replacing a manual per-DC runbook.
- Found a training gap, wrote the ADDS troubleshooting curriculum, and trained 357 technicians across four organisations. Red Flag 17-1 blue-team operator; Superior Performer; Junior Technician of the Year 2017; promoted SSgt in the 94th percentile.

SKILLS

- Application stack:** TypeScript/Node (1.37M tracked LOC), React, Next.js, PostgreSQL (replication, slots, WAL retention, query shape), Redis + Sentinel, AWS, GraphQL, SQL
- Platform and delivery:** ring/canary deploys, progressive rollout, health-gated batch recreate, one-command rollback, podman/Docker, systemd quadlets, k3s, GitHub Actions self-hosted runner fleets, OpenTofu, Cloudflare Workers/Tunnels/Pages
- Reliability:** SLI design, paging-path verification, autoheal and death-watchdog loops, retry and circuit-breaker semantics, single-scheduler egress, incident root-cause converted to gates
- Security:** fail-closed authz, no caller-supplied tenant ids in a decision, secret rotation across 43 verifier sites, vault-backed secrets with audit round-trip, secret scanning and history purge, RBAC
- Languages:** TypeScript, Python, PowerShell, Bash, SQL
- Developer tooling:** code graph and symbol search (awgraph), repo-understanding layer, op-log-aware git for concurrent writers (awgit), self-testing checker framework, tool servers (~1,200 tools)
- Agentic and AI tooling:** multi-agent orchestration, tool-calling harnesses, agent memory and knowledge graphs, evaluation harnesses with blind A/B verdicts, LoRA/fine-tuning, embedding distillation, on-device inference
- Prior depth:** Active Directory at 850K users, HPC tape/SAN/archive, endpoint management at 50K scale, Windows and Linux

CERTIFICATIONS

- Tanium Certified Professional - Endpoint Management (2024-2026)
- Tanium Certified Operator (2022-2026)
- GIAC Certified Windows Security Administrator (GCWN)
- GIAC Certified Enterprise Defender (GCED)
- CompTIA Security+

LINKS

- Who I am (built from this repo) — <https://wizzense.github.io>
- AitherOS / aw* org (61 public repos) — <https://github.com/Aitherium>
- Engineering blog (273 posts) — <https://blog.aitherium.com>
- A code-search embedder 13x smaller that beats its teacher — <https://blog.aitherium.com/code-search-embedder-13x-smaller-beats-its-teacher>

Platform figures re-measured 2026-09-18 on the AitherOS tree; every figure carries the command that produced it.